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# Characterization of $LU$-efficiency and saddle-point criteria for $F$-approximated multiobjective interval-valued variational problems

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ABSTRACT

This paper aims to find a solution of interval uncertainty to multiobjective variational problems. For this, we consider an interval-valued multiobjective variational problem. Then, by using the modified $F$-objective function method, we construct associated interval-valued multiobjective variational problem with the modified $F$-objective functions. We establish a relationship between $LU$-pareto optimal solution of original problem and its associated modified problem by using the concept of $LU$-$F$-convexity and $LU$-$F$-pseudoconvexity. Further, we define $LU$-Lagrange function and its saddle point to discuss the efficient solution of original problem through it. We provide an example to validate our results numerically.

## 1. Introduction

Convexity has a vital role in several aspects of optimization theory like optimality conditions, duality. But all the mathematical models arise in many real world problems likewise decision science, economics, engineering, and so forth, are not necessarily convex. Therefore, many generalizations of a convex function discussed for relaxing the convexity assumption to derive optimality and duality results for such nonconvex optimization problems, in multiobjective optimization sense (see, for example, [1–5], etc.).

Optimization problems having real coefficients are categorized as the deterministic optimization problems. However, the coefficients can be taken as uncertain quantities. If the coefficients of optimization problem are assumed as random variables with known distributions, they are categorized as the stochastic optimization problems. On the other hand, if the coefficients are taken as closed intervals, they will be categorized as interval-valued optimization problems. The specifications of the distributions are more subjective, as many authors invoke the Gaussian distributions for various parameters in the stochastic optimization problems. These specifications may not perfectly match the real problems. Therefore, interval-valued optimization problems may provide an alternative choice for considering the uncertainty into the optimization problems. That is to say, the coefficients in the interval-valued optimization problems are assumed as closed intervals. Several results on interval-valued optimization problems are studied, see for examples, [6–17].

Variational problems, which come from calculus of variation have been widely studied. Hanson [4] first discussed the relationship between optimization problems and variational problems. Later on, many researchers have shown their interest in this specified area. Craven [3] discussed KT-type necessary conditions for multiobjective variational problems by assuming the pseudo and quasi convexity assumptions. In [18], Arana et al. studied various types of duality results for multiobjective variational problems by using pseudoinvexity. For more details, we can study [2,19–23].

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The term multitime was initially introduced in physics by Dirac (1932) and later used in mathematics [24]. The multitime control theory is related to the partial derivatives of dynamical systems and their optimization over multitime is also known as the multidimensional control problems. Multitime control problems have been applied in various fields of science which lead to the exploration in the last few years both from theoretical and applied viewpoints. Methods for solving nonconvex multitime variational problems, as a type of variational problems, in our opinion, remain some unexplored questions for research.

To the best of our knowledge, there is only a few papers devoted on methods which can be used for characterization of solvability in nonconvex multitime variational problems. In [25], the hypothesis for single-time dynamic programming was stretched out to a multitime situation when the development is $m$-dimensional, and the functional includes a path-independent curvilinear integral and clarifies the utilization of multitime dynamic programming in multitime ideal controls.

Considerable attention in the literature has been given recently to devising new methods by the help of which solvability of the original mathematical programming problem is characterized by optimal solutions of its associated problem. These methods have been also used to derive new saddle point criteria for various classes of nonconvex optimization problems. One of such approaches is the modified objective function, was first introduced by Antczak [26] for solving class of differentiable nonconvex vector optimization problems.

However, to the best of our knowledge, there are no studies in the literature which have investigated the methods for proving saddle point criteria for multiobjective interval-valued variational problems and have used them for the characterization of LU-optimality so far. In the paper, we apply the modified objective function approach and develop its associated multiobjective interval-valued variational problems by changing objective function and set up their equality under $F$-convexity on interval-valued functionals. The paper is organized as follows: In Section 2, we recall few fundamentals and consider multiobjective interval-valued variational problems (MIVP). Also, we discuss its solution concept by using partial ordering "$\preceq_{LU}$" on real, closed and bounded intervals. Section 3 discuss $LU$-$F$-saddle-point criteria for the considered problem. In Section 4, by using the idea of modified objective function technique, we construct associated multiobjective interval-valued variational problems by modifying the objective function of the considered problem. We establish the relationship between an efficient solution for the original problem and its associated modified problem. In Section 5, we have shown the relationship between an efficient solution for the original problem and a saddle point in its related modified problem by defining its Lagrange function. Section 6 examine all the recently settled outcomes in the static case. At last, we conclude our work in Section 7.

## 2. Notations and preliminaries

Assume that $X \subset R^n$ and a real interval is represented by $I = [a, b]$. Suppose $\phi : I \times X \times X \mapsto R^n$, and $g : I \times X \times X \mapsto R^m$ are continuously differentiable functions with respect to their arguments. $x : I \mapsto R^n$ is an $n$-dimensional piecewise smooth functions with derivative $\dot{x}(t)$ with respect to time variable $t$. The first order partial derivatives of $\phi^i$ with respect to $x,\ \dot{x}$ denoted by $\phi^i_x,\ \phi^i_{\dot{x}}$ such that $\phi^i_x = \left(\frac{\partial \phi^i}{\partial x_1}, \dots, \frac{\partial \phi^i}{\partial x_n}\right)$ and $\phi^i_{\dot{x}} = \left(\frac{\partial \phi^i}{\partial \dot{x}_1}, \dots, \frac{\partial \phi^i}{\partial \dot{x}_n}\right)$. In the similar manner, $g_x,\ g_{\dot{x}}$ represent the partial derivatives of $g$ with respect to $x,\ \dot{x}$. Let $x = (x_1, x_2) \in R^{2n}$ are any two vectors and $\Lambda_n = \{1, \dots, n\}$ is an index set. Then

$(i)$ $x_1 = x_2 \Leftrightarrow x_{1i} = x_{2i}, i \in \Lambda_n$;

$(ii)$ $x_1 < x_2 \Leftrightarrow x_{1i} < x_{2i}, i \in \Lambda_n$;

$(iii)$ $x_1 \leqq x_2 \Leftrightarrow x_{1i} \leqq x_{1i}, i \in \Lambda_n$;

$(iv)$ $x_1 \leq x_2 \Leftrightarrow x_1 \leqq x_1$ and $x_1 \neq x_2$.

Let $K = \{[I^L, I^U] :\ I^L \leq I^U,\ I^L, I^U \in R\}$ and $R^n_K = \{I \in R^n | I_i \in K,\ i \in \Lambda_n, \}$ where $\Lambda_n = \{1, \dots, n\}$ is an indexed set.

Let $I, J \in K$ and $\alpha \in R$. Then

$$I + J = [I^L + J^L, I^U + J^U],$$
$$I + \alpha = [I^L + \alpha, I^U + \alpha] \text{ and}$$

and

$$I\alpha = \begin{cases} [I^L\alpha, I^U\alpha];\ \alpha \geq 0, \\ [I^U\alpha, I^L\alpha];\ \alpha < 0. \end{cases}$$

It is obvious that $-I = -[I^L, I^U] = [-I^U, -I^L]$.

$$I - J = [I^L, I^U] - [J^L, J^U] = [I^L, I^U] + [-J^U, -J^L] = [I^l - J^U, I^U - J^L].$$

For $I, J \in K$, we use the following conventions of inequalities:

$$I \preceq_{LU} J \Rightarrow I^L \leq J^L,\ I^U \leq J^U;$$
$$I \prec_{LU} J \Rightarrow \begin{cases} I^L < J^L, \\ I^U < J^U, \end{cases} \quad \begin{cases} I^L \leq J^L, \\ I^U < J^U, \end{cases} \quad \begin{cases} I^L < J^L, \\ I^U \leq J^U. \end{cases}$$

Also, let $I, J \in R^n_K$. Then, from [27],

$I \preceq_{LU} J$ iff $I_i \preceq_{LU} J_i,\ i \in \Lambda_n$ and

$I \prec_{LU} J$ iff $I_i \preceq_{LU} J_i,\ i \in \Lambda_n,\ I_h \prec_{LU} J_h$ for at least one index $h$.

**Definition 2.1.** A functional $\psi : I \times X \times X \mapsto R_K$ denoted by $\int_a^b \psi(t, x(t), \dot{x}(t))dt = [\int_a^b \psi^L(t, x(t), \dot{x}(t))dt, \int_a^b \psi^U(t, x(t), \dot{x}(t))dt]$ said to be interval valued function if,

$$\int_a^b \psi^L(t, x(t), \dot{x}(t))dt \leq \int_a^b \psi^U(t, x(t), \dot{x}(t))dt, \ \forall\ x \in X,$$

where $\int_a^b \psi^L(t, x(t), \dot{x}(t))dt, \int_a^b \psi^U(t, x(t), \dot{x}(t))dt$ are lower and upper real end points of integral.

**Definition 2.2.** If $\forall\ x, \dot{x}, u, \dot{u} \in X$, the following inequalities hold:

$(i)$ $F(t, x, \dot{x}, u, \dot{u}; \theta_1 + \theta_2) \leqq F(t, x, \dot{x}, u, \dot{u}; \theta_1) + F(t, x, \dot{x}, u, \dot{u}; \theta_2)$,

$(ii)$ $F(t, x, \dot{x}, u, \dot{u}; a\theta) = aF(t, x, \dot{x}, u, \dot{u}; \theta)$, for any $a \geqq 0,\ \theta_1, \theta_2 \in R$

then the functional $F : I \times X \times X \times X \times X \times R^n \mapsto R$ is said to be sublinear with respect to sixth variable. After seeing the condition (ii), we can write $F(t, x, \dot{x}, u, \dot{u}; 0) = 0$. For sake of simplicity, we use $F(t, x, u; \theta)$ in place of $F(t, x, \dot{x}, u, \dot{u}; \theta)$.

In the paper, we focus on the following problem:

$$\begin{aligned} (\textbf{MIVP})\ \ \text{minimize}\ \ & \left(\int_a^b \psi^1(t, x, \dot{x})dt, \dots, \int_a^b \psi^r(t, x, \dot{x})dt\right) \\ \text{subject to}\ \ & g(t, x, \dot{x}) \leqq 0,\ t \in I, \\ & x(a) = \alpha,\ x(b) = \beta, \end{aligned}$$

where $\int_a^b \psi^i(t, x, \dot{x})dt = [\int_a^b \psi^{iL}(t, x, \dot{x})dt, \int_a^b \psi^{iU}(t, x, \dot{x})dt],\ i \in \Lambda_r = \{1, \dots, r\}$.

Further, $\rhd$ denotes the set of all feasible solutions to (MIVP), i.e.,

$$\rhd = \{x \in X :\ g(t, x, \dot{x}) \leqq 0,\ t \in I,\ x(a) = \alpha,\ x(b) = \beta\}.$$

In order to give the definition of generalized $F$-convexity for interval valued functional, first we give its definition in real valued case [28]. Let $D$ denotes the differential operator.

**Definition 2.3.** If $\forall\ x \in \rhd$

$$\int_a^b \phi(t, x, \dot{x})dt - \int_a^b \phi(t, y, \dot{y})dt \geqq \int_a^b F(t, x, y; \phi_x(t, y, \dot{y}) - D\phi_{\dot{x}}(t, y, \dot{y}))dt$$

then a real valued functional $\int_a^b \phi(t, x, \dot{x})dt$ is known as $F$-convex at $y \in \rhd$.

**Definition 2.4.** An interval-valued functional $\int_a^b f(t, x, \dot{x})dt$ is known as $F$-convex at $y \in \rhd$, if both real ends $\int_a^b f^L(t, x, \dot{x})dt,\ \int_a^b f^U(t, x, \dot{x})dt$ are $F$-convex at $y \in \rhd$.

**Definition 2.5.** A real valued functional $\int_a^b \phi(t, x, \dot{x})dt$ is said to be (strictly) $F$-pseudoconvex at $y \in \rhd$, if for all $x \in \rhd$

$$\int_a^b F(t, x, y; \phi_x(t, y, \dot{y}) - D\phi_{\dot{x}}(t, y, \dot{y}))dt \geqq 0 \Rightarrow \int_a^b \phi(t, x, \dot{x})dt \geqq (>) \int_a^b \phi(t, y, \dot{y})dt$$

or equivalently,

$$\int_a^b \phi(t, x, \dot{x})dt < (\leqq) \int_a^b \phi(t, y, \dot{y})dt \Rightarrow \int_a^b F(t, x, y; \phi_x(t, y, \dot{y}) - D\phi_{\dot{x}}(t, y, \dot{y}))dt < 0.$$

**Definition 2.6.** An interval valued functional $\int_a^b f(t, x, \dot{x})dt$ is said to be $F$-pseudoconvex at $y \in \rhd$, if both real ends $\int_a^b f^L(t, x, \dot{x})dt$, $\int_a^b f^U(t, x, \dot{x})dt$ are $F$-pseudoconvex at $y \in \rhd$.

**Remark 2.1.** From now onwards, we shall use the hypothesis that the sublinear function $F$ vanishes at the same arguments i.e., $F(t, x, x; \cdot) = 0$.

**Definition 2.7.** A feasible point $\bar{x} \in \rhd$ is said to be an $LU$-efficient solution of (MIVP), if there exists no $x \in \rhd$ such that

$$\int_a^b \psi^i(t, x, \dot{x})dt \preceq_{LU} \int_a^b \psi^i(t, \bar{x}, \dot{\bar{x}})dt,\ \forall i \in \Lambda_r,$$
$$\int_a^b \psi^h(t, x, \dot{x})dt \prec_{LU} \int_a^b \psi^h(t, \bar{x}, \dot{\bar{x}})dt,\ \text{for some } h \in \Lambda_r.$$

**Definition 2.8.** A feasible point $\bar{x} \in \rhd$ is said to be a weak $LU$-efficient solution of (MIVP), if there exists no $x \in \rhd$ such that

$$\int_a^b \psi^i(t, x, \dot{x})dt \prec_{LU} \int_a^b \psi^i(t, \bar{x}, \dot{\bar{x}})dt,\ \forall i \in \Lambda_r.$$

Based on the necessary optimality conditions in [29] for single objective interval-valued variational problems, we are writing the same for (MIVP).

**Theorem 2.1.** *Let $\bar{x} \in \rhd$ be a normal LU-efficient solution of (MIVP) Then, there exist $\bar{\lambda}^{iL}, \bar{\lambda}^{iU} \in R,\ i \in \Lambda_r$ and the piecewise smooth multipliers $\bar{\xi}(t) \in R^m_+$ such that*

$$\sum_{i=1}^{r} \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial x}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^{r} \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial x}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)\frac{\partial g}{\partial x}(t,\bar{x},\dot{\bar{x}})$$
$$= D\Big(\sum_{i=1}^{r} \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial \dot{x}}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^{r} \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial \dot{x}}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)\frac{\partial g}{\partial \dot{x}}(t,\bar{x},\dot{\bar{x}})\Big), \tag{1}$$

$$\bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}}) = 0, \tag{2}$$

$$\bar{\lambda}^{iL},\ \bar{\lambda}^{iU} \geq 0,\ \bar{\xi}(t) \geqq 0. \tag{3}$$

**Proof.** Proof of this theorem will follow the same as in [29], hence being omitted.

**Definition 2.9.** An LU-efficient solution $\bar{x} \in \rhd$ is said to be normal for (MIVP) if $\lambda \neq 0$.

According to this definition, without loss of generality we can take $\lambda = 1$.

## 3. *LU*-Saddle-point criteria for (MIVP)

**Definition 3.1.** The Lagrange function for (MIVP) denoted by $L(x,\bar{\lambda},\xi)$ is defined as

$$L(x,\bar{\lambda},\xi) = \int_a^b \Big\{\sum_{i=1}^{r} \bar{\lambda}^{iL}\psi^{iL}(t,x,\dot{x}) + \sum_{i=1}^{r} \bar{\lambda}^{iU}\psi^{iU}(t,x,\dot{x}) + \xi(t)g(t,x,\dot{x})\Big\}dt$$

**Definition 3.2.** A point $(\bar{x},\bar{\lambda},\bar{\xi})$ is said to be saddle-point for the Lagrange function in (MIVP) if, the following two inequalities hold:

$(i)$ $L(\bar{x},\bar{\lambda},\bar{\xi}) \leqq L(x,\bar{\lambda},\bar{\xi}),\ \forall\ x \in \rhd,$

$(ii)$ $L(\bar{x},\bar{\lambda},\xi) \leqq L(\bar{x},\bar{\lambda},\bar{\xi}),\ \forall\ \xi \in R^m_+.$

**Theorem 3.1.** *Let $\bar{x} \in \rhd$ satisfy the conditions (1)–(3) for the Lagrange multipliers $\bar{\lambda}^{iL}, \bar{\lambda}^{iU} \in R;\ \bar{\xi} \in R^m_+$. Assume that the functionals $\int_a^b \psi^{iL}(t,x,\dot{x})dt, \int_a^b \psi^{iU}(t,x,\dot{x})dt$, and $\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt$ are $F$-convex at $\bar{x} \in \rhd$. Then $(\bar{x},\bar{\lambda},\bar{\xi})$ is an LU-saddle point for the Lagrange function in (MIVP).*

**Proof.** Let us assume that the point $(\bar{x},\bar{\lambda},\bar{\xi})$ satisfy the conditions (1)–(3). Now, by the definition of $F$-convexity for the functionals, we have

$$\int_a^b \psi^{iL}(t,x,\dot{x})dt - \int_a^b \psi^{iL}(t,\bar{x},\dot{\bar{x}})dt \geqq \int_a^b F(t,x,\bar{x};\psi^{iL}_x(t,\bar{x},\dot{\bar{x}}) - D\psi^{iL}_{\dot{x}}(t,\bar{x},\dot{\bar{x}}))dt. \tag{4}$$

$$\int_a^b \psi^{iU}(t,x,\dot{x})dt - \int_a^b \psi^{iU}(t,\bar{x},\dot{\bar{x}})dt \geqq \int_a^b F(t,x,\bar{x};\psi^{iU}_x(t,\bar{x},\dot{\bar{x}}) - D\psi^{iU}_{\dot{x}}(t,\bar{x},\dot{\bar{x}}))dt. \tag{5}$$

$$\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt - \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt \geqq \int_a^b F(t,x,\bar{x};\bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}}) - D\bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}}))dt. \tag{6}$$

Multiplying (4), (5) by $\bar{\lambda}^{iL},\ \bar{\lambda}^{iU}$, respectively and adding them over $i$ together with (6), we get

$$\int_a^b \Big(\sum \bar{\lambda}^{iL}\psi^{iL}(t,x,\dot{x}) + \sum \bar{\lambda}^{iU}\psi^{iU}(t,x,\dot{x}) + \bar{\xi}(t)g(t,x,\dot{x})\Big)dt$$
$$- \int_a^b \Big(\sum \bar{\lambda}^{iL}\psi^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum \bar{\lambda}^{iU}\psi^{iU}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})\Big)dt$$
$$\geqq \int_a^b F\Big(t,x,\bar{x};\sum \bar{\lambda}^{iL}\psi^{iL}_x(t,\bar{x},\dot{\bar{x}}) + \sum \bar{\lambda}^{iU}_x\psi^{iU}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)g_x(t,\bar{x},\dot{\bar{x}})$$
$$- D\Big(\sum \bar{\lambda}^{iL}\psi^{iL}_{\dot{x}}(t,\bar{x},\dot{\bar{x}}) + \sum \bar{\lambda}^{iU}\psi^{iU}_{\dot{x}}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)g_{\dot{x}}(t,\bar{x},\dot{\bar{x}})\Big)\Big)dt.$$

Using the necessary optimality condition (1), and sublinearity property, the above inequality becomes

$$\int_a^b \Big(\sum \bar{\lambda}^{iL}\psi^{iL}(t,x,\dot{x}) + \sum \bar{\lambda}^{iU}\psi^{iU}(t,x,\dot{x}) + \bar{\xi}(t)g(t,x,\dot{x})\Big)dt$$
$$\geqq \int_a^b \Big(\sum \bar{\lambda}^{iL}\psi^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum \bar{\lambda}^{iU}\psi^{iU}(t,\bar{x},\dot{\bar{x}}) + \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})\Big)dt,$$

which by the definition of Lagrange function, yields

$$L(x, \bar{\lambda}, \bar{\xi}) \geqq L(\bar{x}, \bar{\lambda}, \bar{\xi}), \ \forall x \in \rhd. \tag{7}$$

Now, from the condition (3) and feasibility of $\bar{x}$, we have

$$\int_a^b \xi(t) g(t, \bar{x}, \dot{\bar{x}}) dt \leqq 0.$$

By using condition (2), the above inequality yields

$$\int_a^b \xi(t) g(t, \bar{x}, \dot{\bar{x}}) dt \leqq \int_a^b \bar{\xi}(t) g(t, \bar{x}, \dot{\bar{x}}) dt.$$

Adding the term $\int_a^b \sum \bar{\lambda}^{iL} \psi^{iL}(t, x, \dot{x}) + \sum \bar{\lambda}^{iU} \psi^{iU}(t, x, \dot{x}) dt$ on both sides of the above inequality, we get

$$\int_a^b \Big( \sum \bar{\lambda}^{iL} \psi^{iL}(t, \bar{x}, \dot{\bar{x}}) + \sum \bar{\lambda}^{iU} \psi^{iU}(t, \bar{x}, \dot{\bar{x}}) + \xi(t) g(t, \bar{x}, \dot{\bar{x}}) \Big) dt$$
$$\leqq \int_a^b \Big( \sum \bar{\lambda}^{iL} \psi^{iL}(t, \bar{x}, \dot{\bar{x}}) + \sum \bar{\lambda}^{iU} \psi^{iU}(t, \bar{x}, \dot{\bar{x}}) + \bar{\xi}(t) g(t, \bar{x}, \dot{\bar{x}}) \Big) dt,$$

which again by the definition of the Lagrange function, yields

$$L(\bar{x}, \bar{\lambda}, \xi) \leqq L(\bar{x}, \bar{\lambda}, \bar{\xi}), \ \forall \xi \in R^m_+. \tag{8}$$

Thus, (7) and (8) show that the point $(\bar{x}, \bar{\lambda}, \bar{\xi})$ is a $LU$-saddle point for the Lagrange function in (MIVP). □

## 4. Sufficient condition for multiobjective interval-valued variational problems through $F$-modified objective function approach

Let $\bar{x}$ be an arbitrary given feasible solution (MIVP). Then the multiobjective variational problem (MIVP$_F(\bar{x})$) with the modified $LU$-$F$-objective function corresponding to (MIVP) is constructed as follows:

$$\begin{aligned} (\mathbf{MIVP}_F(\bar{x})) \quad & \text{minimize } (H^1, \dots, H^r) \\ & \text{subject to} \quad g(t, x, \dot{x}) \leqq 0, \\ & \qquad x(a) = \alpha, \ x(b) = \beta, \end{aligned}$$

where

$$H^i(x) = [\int_a^b F(t, x, \bar{x}; \psi_x^{iL}(t, \bar{x}, \dot{\bar{x}}) - D\psi_{\dot{x}}^{iL}(t, \bar{x}, \dot{\bar{x}})) dt, \int_a^b F(t, x, \bar{x}; \psi_x^{iU}(t, \bar{x}, \dot{\bar{x}}) - D\psi_{\dot{x}}^{iU}(t, \bar{x}, \dot{\bar{x}})) dt],$$

$\psi^{iL}$, $\psi^{iU}$ and $g$ are previously mentioned in Section 2.

**Remark 4.1.** It is clear that the feasible set of (MIVP$_F(\bar{x})$) is the same as that of (MIVP).

**Definition 4.1.** A feasible solution $\bar{x} \in \rhd$ is said to be an $LU$-efficient solution of (MIVP$_F(\bar{x})$), if there exists no $y \in \rhd$ such that

$$H^i(y) \preceq_{LU} H^i(\bar{x}), \ \forall \ i \in \Lambda_r,$$
$$H^h(y) \prec_{LU} H^h(\bar{x}), \ \text{for at least one } h \in \Lambda_r.$$

We present the equivalence between (MIVP) and (MIVP$_F(\bar{x})$) through $LU$-efficiency under weaker hypotheses.

**Theorem 4.1.** *Let $\bar{x}$ be an $LU$-efficient solution to (MIVP$_F(\bar{x})$). Assume that $\int_a^b (\bar{\lambda}^L \psi^L(t, x, \dot{x}) + \bar{\lambda}^U \psi^U(t, x, \dot{x})) dt$ is strictly $LU$-$F$-pseudoconvex at $\bar{x} \in \rhd$ on $\rhd$. Then, $\bar{x}$ is also an $LU$-efficient solution to (MIVP).*

**Proof.** Contrarily, we assume that $\bar{x}$ is not an $LU$-efficient solution of (MIVP). Then for the existence of $y \in \rhd$, the inequalities

$$\Big[\int_a^b \psi^{iL}(t, y, \dot{y}) dt, \int_a^b \psi^{iU}(t, y, \dot{y}) dt\Big] \preceq_{LU} \Big[\int_a^b \psi^{iL}(t, \bar{x}, \dot{\bar{x}}) dt, \int_a^b \psi^{iU}(t, \bar{x}, \dot{\bar{x}}) dt\Big], \ \forall \ i \in \Lambda_r,$$
$$\Big[\int_a^b \psi^{iL}(t, y, \dot{y}) dt, \int_a^b \psi^{iU}(t, y, \dot{y}) dt\Big] \prec_{LU} \Big[\int_a^b \psi^{iL}(t, \bar{x}, \dot{\bar{x}}) dt, \int_a^b \psi^{iU}(t, \bar{x}, \dot{\bar{x}}) dt\Big],$$
$$\text{for at least one } h \in \Lambda_r$$

hold. By using the partial order relation on interval analysis, the above relation can be written as

$$\int_a^b \psi^{iL}(t, y, \dot{y}) dt \leqq \int_a^b \psi^{iL}(t, \bar{x}, \dot{\bar{x}}) dt, \ \int_a^b \psi^{iU}(t, y, \dot{y}) dt \leqq \int_a^b \psi^{iU}(t, \bar{x}, \dot{\bar{x}}) dt, \ \forall \ i \in \Lambda_r$$

and

$$\begin{cases}\int_a^b \psi^{hL}(t,y,\dot{y})dt < \int_a^b \psi^{hL}(t,\bar{x},\dot{\bar{x}})dt\\ \int_a^b \psi^{hU}(t,y,\dot{y})dt < \int_a^b \psi^{hU}(t,\bar{x},\dot{\bar{x}})dt,\end{cases} \qquad \begin{cases}\int_a^b \psi^{hL}(t,y,\dot{y})dt \leqq \int_a^b \psi^{hL}(t,\bar{x},\dot{\bar{x}})dt\\ \int_a^b \psi^{hU}(t,y,\dot{y})dt < \int_a^b \psi^{hU}(t,\bar{x},\dot{\bar{x}})dt,\end{cases}$$

$$\begin{cases}\int_a^b \psi^{hL}(t,y,\dot{y})dt < \int_a^b \psi^{hL}(t,\bar{x},\dot{\bar{x}})dt\\ \int_a^b \psi^{hU}(t,y,\dot{y})dt \leqq \int_a^b \psi^{hU}(t,\bar{x},\dot{\bar{x}})dt,\end{cases} \quad \text{for at least one h } \in \Lambda_r.$$

Multiplying the above inequalities by $\bar{\lambda}^{iL},\ \bar{\lambda}^{iU},\ \forall\ i \in \Lambda_r$ and $\bar{\lambda}^{hL},\ \bar{\lambda}^{hU},$ for at least one $h \in \Lambda_r$ and adding them

$$\int_a^b \sum(\lambda^{iL}\psi^{iL}(t,y,\dot{y}) + \lambda^{iU}\psi^{iU}(t,y,\dot{y}))dt \leqq \int_a^b \sum(\lambda^{iL}\psi^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi^{iU}(t,\bar{x},\dot{\bar{x}}))dt,\ \forall i \in \Lambda_r, \tag{9}$$

$$\int_a^b \sum(\lambda^{hL}\psi^{hL}(t,y,\dot{y}) + \lambda^{hU}\psi^{hU}(t,y,\dot{y}))dt < \int_a^b \sum(\lambda^{hL}\psi^{hL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{hU}\psi^{hU}(t,\bar{x},\dot{\bar{x}}))dt,\ \text{for at least one } h \in \Lambda_r. \tag{10}$$

Since $\int_a^b(\bar{\lambda}^L\psi^L(t,x,\dot{x}) + \bar{\lambda}^U\psi^U(t,x,\dot{x}))dt$ is strictly $LU$-$F$ pseudoconvex at $\bar{x}$, therefore, (9) and (10) implies that

$$\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt \leqq 0,\ \forall i \in \Lambda_r,$$

$$\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt < 0,\ \text{for at least one } h \in \Lambda_r.$$

Since the sublinear functional $F$ vanishes with respect to the same argument, the above inequalities become

$$\begin{aligned}&\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt\\ &\leqq \int_a^b F\Big(t,\bar{x},\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt,\ \forall i \in \Lambda_r,\\ &\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt\\ &< \int_a^b F\Big(t,\bar{x},\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt,\ \text{for at least one } h \in \Lambda_r,\end{aligned}$$

which shows that $y$ is an $LU$-efficient solution of (MIVP$_F(\bar{x})$), which is the contradiction. This completes the proof. □

Now, we prove the converse of Theorem 4.1.

**Theorem 4.2.** *Suppose $\bar{x}$ is an $LU$-efficient solution to (MIVP at which the conditions (1)–(3) are satisfied for the multipliers $\bar{\lambda}^{iL}, \bar{\lambda}^{iU} \in R$; $\bar{\xi}(t) \in R^m$. Then $\bar{x}$ is also an $LU$-efficient solution to (MIVP$_F(\bar{x})$) provided $\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt$ is $F$-convex at $\bar{x} \in \triangleright$ on $\triangleright$.*

**Proof.** Suppose $\bar{x}$ satisfies conditions (1)–(3) together with the Lagrange multipliers $\bar{\lambda},\ \bar{\xi}(t)$. On the contrary, suppose $\bar{x}$ is not an efficient solution of (MIVP$_F(\bar{x})$). Then, there exists $y \in \triangleright$ such that

$$\Big[\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) - D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt, \int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) - D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt\Big]$$
$$\leq_{LU} \Big[\int_a^b F(t,\bar{x},\bar{x};\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))dt, \int_a^b F(t,\bar{x},\bar{x};\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) - D\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))dt\Big],\ \forall i \in \Lambda_r,$$

$$\Big[\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt,\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt\Big]$$
$$\prec_{LU}\Big[\int_a^b F(t,\bar{x},\bar{x};\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))dt,$$
$$\int_a^b F(t,\bar{x},\bar{x};\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))dt\Big],\ \text{for at least one } h\in\Lambda_r.$$

Using the hypothesis $F(t,\bar{x},\bar{x};.)=0$ in the above inequality, we have

$$\Big[\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt,\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt\Big]$$
$$\preceq_{LU} 0,\ \forall i\in\Lambda_r,$$
$$\Big[\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt,\int_a^b F(t,y,\bar{x};\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})-D\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}))dt\Big]$$
$$\prec_{LU} 0,\ \text{for at least one } h\in\Lambda_r.$$

Since $\bar{\lambda}^{iL},\bar{\lambda}^{iU}\geq 0$, therefore, multiplying the above inequality by these multipliers and adding them, we get

$$\int_a^b F\Big(t,y,\bar{x};\sum\Big(\bar{\lambda}^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})+\bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$-D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}})+\bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt<0. \tag{11}$$

Since $\int_a^b\bar{\xi}(t)g(t,x,\dot{x})dt$ is $F$-convex, therefore, by the definition, we have

$$\int_a^b F(t,y,\bar{x};\bar{\xi}(t)g_x(t,\bar{x},\dot{\bar{x}})-D(\bar{\xi}(t)g_{\dot{x}}(t,\bar{x},\dot{\bar{x}})))dt\leqq\int_a^b\bar{\xi}(t)g(t,y,\dot{y})dt-\int_a^b\bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt$$

Since $y\in\rhd$, therefore, by the feasibility of $y$ and necessary optimality condition (2) and (3), the above inequality reduces to

$$\int_a^b F(t,y,\bar{x};\bar{\xi}(t)g_x(t,\bar{x},\dot{\bar{x}})-D(\bar{\xi}(t)g_{\dot{x}}(t,\bar{x},\dot{\bar{x}})))dt\leqq 0. \tag{12}$$

Adding (11) and (12), we get

$$\int_a^b F\Big(t,y,\bar{x};\Big(\sum\bar{\lambda}^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})+\sum\bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})+\bar{\xi}(t)g_x(t,\bar{x},\dot{\bar{x}})$$
$$-D(\sum\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}})+\sum\bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}})+\bar{\xi}(t)g_{\dot{x}}(t,\bar{x},\dot{\bar{x}})))\Big)dt<0,$$

which contradicts the necessary condition (1). This completes the proof. □

## 5. $LU$-$F$-Saddle-point criteria for (MIVP$_F(\bar{x})$)

In this section, under generalized invexity hypotheses, we establish the equivalence between an $LU$-efficient solution of (MIVP) and an $LU$-$F$-saddle-point of the Lagrangian in (MIVP$_F(\bar{x})$).

By keeping the notion of Lagrange function for variational problems [30,31], we give the following definitions for the Lagrange function and saddle-point in (MIVP$_F(\bar{x})$).

**Definition 5.1.** The Lagrange function for (MIVP$_F(\bar{x})$) is denoted by $L_F(x,\bar{x},\bar{\xi})$ and is defined as

$$L_F(x,\bar{\lambda},\bar{\xi})=\int_a^b F\Big(t,x,\bar{x};\sum_{i=1}^r\bar{\lambda}^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}})+\sum_{i=1}^r\bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$-D\Big\{\sum_{i=1}^r\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}})+\sum_{i=1}^r\bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}})\Big\}\Big)dt+\int_a^b\bar{\xi}(t)g(t,x,\dot{x})dt.$$

**Definition 5.2.** A point $(\bar{x},\bar{\lambda},\bar{\xi})\in\rhd\times R^{2r}\times R^m$ is said to be an $LU$-$F$-saddle-point of the Lagrangian in (MIVP$_F(\bar{x})$), if

$(i)$ $L_F(\bar{x},\bar{\lambda},\xi)\leqq L(\bar{x},\bar{\lambda},\bar{\xi}),\ \forall\ \xi\in R_+^m,$

$(ii)$ $L_F(\bar{x},\bar{\lambda},\bar{\xi})\leqq L(x,\bar{\lambda},\bar{\xi}),\ \forall\ x\in\rhd$

hold.

**Theorem 5.1.** *Let $\bar{x}\in\rhd$ at which the functional $\int_a^b(\bar{\lambda}^L\psi^L(t,x,\dot{x})+\bar{\lambda}^U\psi^U(t,x,\dot{x}))dt$ is $F$-pseudoconvex. If $(\bar{x},\lambda,\xi)$ is an $LU$-$F$-saddle point for the Lagrangian in (MIVP$_F(\bar{x})$) then it is an $LU$-efficient solution of (MIVP).*

**Proof.** Since $\bar{x}$ is an $LU$-$F$-saddle point in (MIVP$_F(\bar{x})$), therefore, from the inequality $(i)$ in the definition of saddle-point, we have

$$L_F(\bar{x},\bar{\lambda},\xi(t)) \leqq L_F(\bar{x},\bar{\lambda},\bar{\xi}(t)),\ \forall \xi \in R^m_+,$$

which by Lagrangian's definition, yields

$$\begin{aligned}
&\int_a^b F\Big(t,x,\bar{x};\sum_{i=1}^r \bar{\lambda}^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) \\
&\qquad - D\Big\{\sum_{i=1}^r \bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}})\Big\}\Big)dt + \int_a^b \xi(t)g(t,\bar{x},\dot{\bar{x}})dt \\
&\leqq \int_a^b F\Big(t,x,\bar{x};\sum_{i=1}^r \bar{\lambda}^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) \\
&\qquad - D\Big\{\sum_{i=1}^r \bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}})\Big\}\Big)dt + \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt.
\end{aligned}$$

Since $\xi \in R^m_+$, therefore, let us take $\xi = 0$ in the above inequality, we get

$$\int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt \geqq 0. \tag{13}$$

Since $\bar{x} \in \rhd$, therefore from the feasibility together with $\bar{\xi}(t) \in R^m_+$

$$\int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt \geqq 0. \tag{14}$$

Combining (13) and (14), we get

$$\int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt = 0. \tag{15}$$

Contrarily, assume that $\bar{x}$ is not an efficient solution of (MIVP). Then, for the existence of $y \in \rhd$

$$\begin{aligned}
&\Big[\int_a^b \psi^{iL}(t,y,\dot{y})dt, \int_a^b \psi^{iU}(t,y,\dot{y})dt\Big] \preceq_{LU} \Big[\int_a^b \psi^{iL}(t,\bar{x},\dot{\bar{x}})dt, \int_a^b \psi^{iU}(t,\bar{x},\dot{\bar{x}})dt\Big],\ \forall\ i \in \Lambda_r, \\
&\Big[\int_a^b \psi^{iL}(t,y,\dot{y})dt, \int_a^b \psi^{iU}(t,y,\dot{y})dt\Big] \prec_{LU} \Big[\int_a^b \psi^{iL}(t,\bar{x},\dot{\bar{x}})dt, \int_a^b \psi^{iU}(t,\bar{x},\dot{\bar{x}})dt\Big], \\
&\qquad\qquad \text{for at least one } h \in \Lambda_r
\end{aligned}$$

hold. Multiplying the above inequalities by $\bar{\lambda}^{iL}, \bar{\lambda}^{iU}$ and adding them

$$\begin{aligned}
&\int_a^b \sum(\lambda^{iL}\psi^{iL}(t,y,\dot{y}) + \lambda^{iU}\psi^{iU}(t,y,\dot{y}))dt \\
&\qquad < \int_a^b \sum(\lambda^{iL}\psi^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi^{iU}(t,\bar{x},\dot{\bar{x}}))dt,\ \forall i \in \Lambda_r.
\end{aligned}$$

Since $\int_a^b(\bar{\lambda}^L\psi^L(t,x,\dot{x}) + \bar{\lambda}^U\psi^U(t,x,\dot{x}))dt$ is $F$-pseudoconvex, therefore, the above inequality yields

$$\begin{aligned}
&\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) \\
&\qquad - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt < 0.
\end{aligned}$$

Since $F(t,\bar{x},\bar{x};\cdots) = 0$, therefore, by combining the above inequality together with (15), and feasibility of $y$ we find

$$\begin{aligned}
&\int_a^b F\Big(t,y,\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) \\
&\qquad - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \bar{\xi}(t,y,\dot{y})dt \\
&< \int_a^b F\Big(t,\bar{x},\bar{x};\sum\Big(\lambda^{iL}\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}}) \\
&\qquad - D(\lambda^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \lambda^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \bar{\xi}(t,\bar{x},\dot{\bar{x}})dt,
\end{aligned}$$

which again by Lagrangian's definition, yields

$$L_F(y,\bar{\lambda},\bar{\xi}(t)) < L_F(\bar{x},\bar{\lambda},\bar{\xi}(t)),\ \text{for any } y \in \rhd.$$

This is a contradiction to the second inequality of saddle-point.

Now, we justify Theorems 4.2 and 5.1 numerically.

**Example 5.1.** Let $X = [0, \infty)$, $I = [0, 1]$. Consider the problem:

$$\text{(MIVP1)} \quad \text{minimize} \quad \Big(\Big[\int_0^1 (x^2(t) + \arctan x(t))dt, \ \int_0^1 (x(t) + x^3(t) + 2)dt\Big], \\ \Big[\int_0^1 (-\arccos x(t) - x^2(t) - x^3(t))dt, \ \int_0^1 (-x(t) - x^3(t))dt\Big]\Big) \\ \text{subject to} \quad g(t, x, \dot{x}) = x^2(t) - x(t) \leqq 0.$$

$\rhd = \{x \in X : 0 \leqq x(t) \leqq 1, \ t \in I\}$. The feasible point $\bar{x} = 0 \in \rhd$. Let $F(t, x, \bar{x}; a) = a(x(t) - \bar{x}(t))$. It is obvious that $F(t, \bar{x}, \bar{x}; \cdot) = 0$. $\int_0^1 \bar{\xi}(t)g(t, x, \dot{x})dt$ is $F$-convex at $\bar{x} = 0$ because

$$\int_0^1 \bar{\xi}(t)g(t, x, \dot{x})dt - \int_0^1 \bar{\xi}(t)g(t, \bar{x}, \dot{\bar{x}})dt \\ - \int_0^1 F(t, x, \bar{x}; \bar{\xi}(t)g_x(t, \bar{x}, \dot{\bar{x}}) - D(\bar{\xi}(t)g_{\dot{x}}(t, \bar{x}, \dot{\bar{x}})))dt = \int_0^1 x^2(t)dt \geqq 0, \ \forall \ x \in \rhd.$$

Similarly, we can verify $F$-pseudoconvexity of $\int_a^b (\bar{\lambda}^L \psi^L(t, x, \dot{x}) + \bar{\lambda}^U \psi^U(t, x, \dot{x}))dt$ at $\bar{x} \in \rhd$. Also, conditions (1)–(3) are satisfied at $\bar{x} = 0$ together with the multipliers $\bar{\lambda}^{1L} = \bar{\lambda}^{2L} = 1$, $\bar{\lambda}^{1U} = 2, \bar{\lambda}^{2U} = 1; \bar{\xi}(t) = 1$.

The associated $F$-approximated problem (MIVP$_F(\bar{x})$) is constructed as :

$$\text{(MIVP1}_F(\bar{x})) \quad \text{minimize} \quad \Big(\Big[\int_0^1 (x(t))dt, \ \int_0^1 (x(t))dt\Big], \\ \Big[\int_0^1 (-x(t))dt, \ \int_0^1 (-x(t))dt\Big]\Big) \\ \text{subject to} \quad g(t, x, \dot{x}) = x^2(t) - x(t) \leqq 0.$$

Now, we have to present that $(\bar{x}, \bar{\lambda}, \bar{\xi}) = (0, ([1, 2], [1, 1]), 1)$ is a $LU$-$F$-saddle point for the Lagrangian in (MIVP$_F(\bar{x})$). Lagrangian for (MIVP$_F(\bar{x})$) is defined as

$$L_F(x, \bar{\lambda}, \xi) = \int_0^1 (x(t) + \xi(t)(x^2(t) - x(t)))dt.$$

We have

$$L_F(\bar{x}, \bar{\lambda}, \xi) - L_F(\bar{x}, \bar{\lambda}, \bar{\xi}) = 0, \ \forall \xi \in R^m_+, \\ L_F(\bar{x}, \bar{\lambda}, \bar{\xi}) - L_F(x, \bar{\lambda}, \xi) = -\int_0^1 x^2(t)dt \leqq 0, \ \forall x \in \rhd.$$

Thus, we conclude that $(0, ([1, 2], [1, 1]), 1)$ is a saddle-point for (MIVP$_F(\bar{x})$). Since all the hypotheses of Theorem 5.1 are satisfied at $\bar{x} = 0$, therefore, $\bar{x} = 0$ is an $LU$-efficient solution of (MIVP1). Further, we can also verify Theorem 4.2 from this example also.

**Theorem 5.2.** *Let $\bar{x} \in \rhd$ be an $LU$-efficient solution of (MIVP) at which the conditions (1)–(3) are satisfied together with the multipliers $\bar{\lambda}$, $\bar{\xi}(t)$. Then, $(\bar{x}, \bar{\lambda}, \bar{\xi}) \in \rhd \times R^{2r} \times R^m_+$ is an $LU$-$F$-saddle point for (MIVP$_F(\bar{x})$) provided $\int_a^b \bar{\xi}(t)g(t, x, \dot{x})dt$ is $F$-convex at $\bar{x} \in \rhd$.*

**Proof.** Since $\bar{x}$ is an $LU$-efficient solution of (MIVP), therefore, conditions (1)–(3) are satisfied together with the multipliers $\bar{\lambda}, \bar{\xi}(t)$. Therefore, from condition (1), we have

$$\sum_{i=1}^r \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial x}(t, \bar{x}, \dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial x}(t, \bar{x}, \dot{\bar{x}}) + \bar{\xi}(t)\frac{\partial g}{\partial x}(t, \bar{x}, \dot{\bar{x}}) \\ = D\Big(\sum_{i=1}^r \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}}) + \bar{\xi}(t)\frac{\partial g}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}})\Big).$$

i.e.,

$$\sum_{i=1}^r \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial x}(t, \bar{x}, \dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial x}(t, \bar{x}, \dot{\bar{x}}) \\ - D\Big(\sum_{i=1}^r \bar{\lambda}^{iL} \frac{\partial \psi^{iL}}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}}) + \sum_{i=1}^r \bar{\lambda}^{iU} \frac{\partial \psi^{iU}}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}})\Big) \\ = D\Big(\bar{\xi}(t)\frac{\partial g}{\partial \dot{x}}(t, \bar{x}, \dot{\bar{x}})\Big) - \bar{\xi}(t)\frac{\partial g}{\partial x}(t, \bar{x}, \dot{\bar{x}}). \tag{16}$$

Let $x \in \rhd$ be an arbitrary feasible point. Since $\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt$ is $F$-convex at $\bar{x}$, therefore, by definition of $F$-convexity

$$\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt - \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt$$
$$\geqq \int_a^b F(t,x,\bar{x};\bar{\xi}(t)g_x(t,\bar{x},\dot{\bar{x}}) - D(\bar{\xi}(t)g_{\dot{x}}(t,\bar{x},\dot{\bar{x}})))dt,$$

which by using (16) yields

$$\int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt - \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt$$
$$\geqq -\int_a^b F\Big(t,x,\bar{x};\sum_{i=1}^r\Big(\bar{\lambda}^{iL}\partial\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$- D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt.$$

Since $F(t,\bar{x},\bar{x};.) = 0$, therefore, we have

$$\int_a^b F\Big(t,x,\bar{x};\sum_{i=1}^r\Big(\bar{\lambda}^{iL}\partial\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$- D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \bar{\xi}(t)g(t,x,\dot{x})dt$$
$$\geqq \int_a^b F\Big(t,\bar{x},\bar{x};\sum_{i=1}^r\Big(\bar{\lambda}^{iL}\partial\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$- D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt,$$

which by definition of Lagrange function, yields

$$L_F(\bar{x},\bar{\lambda},\bar{\xi}) \leqq L(x,\bar{\lambda},\bar{\xi}),\ \forall\ x \in \rhd. \tag{17}$$

Now, using the feasibility of $\bar{x}$ together with the multiplier $\xi(t) \in R_+^m$

$$\int_a^b \xi(t)g(t,\bar{x},\dot{\bar{x}})dt \leqq 0,$$

which on using condition (2)

$$\int_a^b \xi(t)g(t,\bar{x},\dot{\bar{x}})dt \leqq \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt.$$

Again using the hypothesis, $F(t,\bar{x},\bar{x};\cdot) = 0$ in the above inequality

$$\int_a^b F\Big(t,\bar{x},\bar{x};\sum_{i=1}^r\Big(\bar{\lambda}^{iL}\partial\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$- D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \xi(t)g(t,\bar{x},\dot{\bar{x}})dt$$
$$\leqq \int_a^b F\Big(t,\bar{x},\bar{x};\sum_{i=1}^r\Big(\bar{\lambda}^{iL}\partial\psi_x^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_x^{iU}(t,\bar{x},\dot{\bar{x}})$$
$$- D(\bar{\lambda}^{iL}\psi_{\dot{x}}^{iL}(t,\bar{x},\dot{\bar{x}}) + \bar{\lambda}^{iU}\psi_{\dot{x}}^{iU}(t,\bar{x},\dot{\bar{x}}))\Big)\Big)dt + \int_a^b \bar{\xi}(t)g(t,\bar{x},\dot{\bar{x}})dt,$$

which, by Lagrangian's definition, yields

$$L_F(\bar{x},\bar{\lambda},\xi) \leqq L(\bar{x},\bar{\lambda},\bar{\xi}),\ \forall\ \xi \in R_+^m. \tag{18}$$

Thus, (17) and (18) show that $(\bar{x},\bar{\lambda},\bar{\xi})$ is an $LU$-$F$-saddle point for (MIVP$_F(\bar{x})$). □

## 6. Static case

If the time dependent variable is dropped between the problems (MIVP) and (MIVP$_F(\bar{x})$), then the following pair of the problems are obtained.

$$(\textbf{SMIVP})\quad \text{minimize} \quad \big([\psi^{1L}(x),\psi^{1U}(x)],\ldots,[\psi^{rL}(x),\psi^{rU}(x)]\big)$$
$$\text{subject to} \quad g(x) \leqq 0.$$

$$(\mathbf{SMIVP}_F(\bar{x})) \quad \text{minimize} \left( [F(x,\bar{x};\psi_x^{1L}(\bar{x})), F(x,\bar{x};\psi_x^{1U}(\bar{x}))], \ldots, [F(x,\bar{x};\psi_x^{rL}(\bar{x})), F(x,\bar{x};\psi_x^{rU}(\bar{x}))] \right)$$
$$\text{subject to} \qquad g(x) \leqq 0.$$

All the results can be establish, which are previously mentioned in the paper.

## 7. Conclusion

We have utilized the modified $F$-objective function approach for uncertainty existing in real world problems, which are modeled into interval-valued multiobjective variational problems. Therefore, the associated $F$-modified interval valued multiobjective variational problems is constructed by modifying the objective functions of the original problems. Then, we have established the equivalence of an $LU$-efficient solution between these problems. Thereafter, we have defined the $LU$-$F$-saddle point for the Lagrange function in (MIVP$_F(\bar{x})$) and also establish the equivalence between $LU$-efficiency and $LU$-$F$-saddle point between these pair of problems. We have supported our outcomes numerically. In addition, we have likewise talked about similar outcomes in static case. The above outcomes should be possible in multitime case which will situate the future undertaking of the creators.

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